

GIANLUIGI VITALE

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EDUCATION

B.Sc. in Computer Engineering — Universitas Mercatorum, Rome, Italy **Expected March 2027**

GPA: 29.35/30 (Italian scale) · 3.87/4.0 (WES-converted) — all coursework completed; thesis in progress

Thesis: DSProfiler — Static Analysis and Machine Learning for Automated Detection of Data Structure Performance Anti-patterns in Python

Supervisor: Prof. Rocco Pietrini

Full Stack Web Development Bootcamp — Boolean SRL (700 hours)

Jan.–Jun. 2020

PUBLICATIONS

G. Vitale, "DriftBench: Measuring and Predicting Infrastructure Drift in LLM Serving Systems," *Proceedings of MLSys 2026*, Bellevue, WA, May 2026. Oral presentation.

Paper: https://proceedings.mlsys.org/paper_files/paper/2026/hash/ea0b5818ae9255ee1fb1e3b4442d2ffe-Abstract-Conference.html

Artifact (ACM badges: Available, Functional, Reproduced): DOI 10.5281/zenodo.19361066

Talk: <https://youtu.be/KnsVj5DdlEo>

UNDER REVIEW

G. Vitale, "Decomposing One Professional-Framing Pipeline: Which Components Shift LLM Safety Boundaries?," *NeurIPS 2026, main track*. Under review.

G. Vitale, "Natural Language Is Flatland for Machine Intelligence: AI Systems Should Decouple Their Reasoning Substrate from Human Language," *NeurIPS 2026, position paper track*. Under review.

RESEARCH INTERESTS

Machine learning systems, LLM serving infrastructure, accelerator portability, TPU kernels and numerical behavior, ML benchmarking, inference reliability, and LLM safety evaluation.

PROFESSIONAL EXPERIENCE

Systems Engineer — Archivio di Stato di Pistoia (Italian State Archive), Italian Ministry of Culture **Apr. 2024–Present**

Designed and deployed a retrieval-augmented generation system over 10,000 internal administrative documents, delivering source-cited answers to staff queries. Built an AI-assisted public consultation interface giving staff and citizens natural-language access to the full inventory database.

Web Developer — Informamuse S.R.L.

Oct. 2020–Apr. 2021

Developed websites and web applications using HTML, CSS, JavaScript, PHP, Laravel, Vue.js, and MySQL. Managed projects end-to-end from design through deployment.

ACADEMIC SERVICE

Program Committee Member — MLSys 2027

2026–2027

Reviewing submissions for the MLSys 2027 technical program.

Ethics Reviewer — NeurIPS 2026

2026

Reviewed submissions for ethical concerns, potential societal harms, research-integrity risks, and compliance with the conference's ethics guidelines.

Artifact Evaluation Committee Member — *SOSP 2026*

2026

Evaluated research artifacts in memory-efficient model serving and memory-safety analysis for GPU kernels in LLM inference systems. Assessed availability, functionality, reproducibility, and consistency with the associated papers' experimental claims.

Artifact Evaluation Committee Member — *MLSys 2026*

2026

Evaluated four accepted submissions spanning attention-kernel optimization, ML profiling infrastructure, GPU memory disaggregation, and distributed inference. Reproduced experimental results on rented 8×H100 clusters (RunPod) and AMD MI350X infrastructure (AMD University Program), assessing availability, functionality, and consistency with each paper's claims.

RESEARCH EXPERIENCE

- **DriftBench and TPU validation:** Built a measurement and prediction framework spanning 236,985 prompt-response pairs across 105 configurations. The Portability Risk Index generalizes to unseen hardware ($R^2 = 0.909$). Extended the result to TPU v5e/v6e, where predictability held across a new numerical path ($R^2 = 0.72-0.96$).
- **DeepSeek V4-Flash on TPU:** Completed a 32-chip v4-64 bring-up, validating every attention mode with both prefill and decode operational. Equivalent support has since landed in vLLM TPU inference through parallel development in the wider ecosystem.
- **DSA kernel for TPU:** Developed a working implementation that is bit-exact against the reference through 8K tokens and runs through 256K. Work in progress; it is the foundation of the GLM-5.2 port.
- **Bit-level TPU MXU characterization:** Characterized TPU v5e/v6e numerical behavior and built a CPU simulator that reproduces real Llama-3.1-8B matrix multiplications bit-for-bit on a native pass, enabling deterministic study of TPU arithmetic.
- **Long-context genomics with CNR Italy:** Collaborating on whether long-context DNA models encode linkage disequilibrium, combining ML systems work with genomics expertise at Italy's National Research Council.
- **Factorial ablation of jailbreak framing:** Ran approximately 31,900 trials across nine LLMs, isolating the independent contributions of persona framing, terminology substitution, and moral justification to safety-boundary shifts.

TECHNICAL SKILLS

Languages: Python, JavaScript, PHP, SQL, HTML/CSS, Bash

Frameworks: PyTorch, JAX, Pallas, Hugging Face Transformers

ML/Systems: vLLM, tpu-inference, SGLang, TensorRT-LLM, CUDA, Docker

Infrastructure: Google Cloud TPU (v4, v5e, v6e), Linux administration, networking, cloud GPU provisioning (RunPod, AMD HPC/Chameleon Cloud), CI/CD

Research Tools: LaTeX, Git, Jupyter, scipy, statsmodels

COMPUTATIONAL RESOURCES

- Google Cloud TPU Research Cloud (TRC) - active since May 2026. Current 320-chip allocation: 32 on-demand v4 plus 288 spot chips across v4, v5e, and v6e in US and EU zones.
- AMD University Program account (active since March 2026)
- Chameleon Cloud allocation (2,000 SUs)
- RunPod GPU cluster experience (H100, H200, B200, MI300X configurations)

LANGUAGES

Italian (native), **English** (C1 — academic proficiency, scientific publications)

CONTACTS & PROFILES

Google Scholar: <https://scholar.google.com/citations?hl=en&user=x4K6NWUAAAAJ> | ORCID: <https://orcid.org/0009-0007-6740-2695>

OpenReview: https://openreview.net/profile?id=~Gianluigi_Vitale1 | Portfolio: <https://gianluigivitale.github.io/portfolio/>

GitHub: <https://github.com/GianluigiVitale> | LinkedIn: <https://www.linkedin.com/in/gianluigi-vitale615>